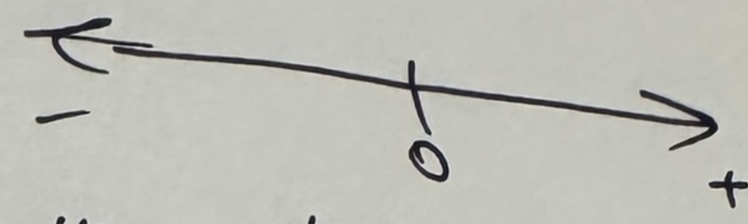

3. Derive the Distribution of the option value, ignoring interest and commissions

option = max(x − 25, 0)
If stock ≤ 25 → value = 0
If stock > 25 ⇒ subtract 25

stock	option value
20-25	0
26	1
27	2
28	3
29	4
30	5

Probability stock ≤ 25:

0.05 + 0.06 + 0.07 + .08 + 0.09 + .12 = 0.47

So:

Option value	probability
0	0.47
1	.15
2	.13
3	.10
4	.10
5	0.05

✓ Q3

4.) What is the probability that the option will have positive value?

Positive value happens when stock > 25
↓
greater

Probabilities for 26-30:

.15 + .13 + .10 + .10 + 0.05 = $\boxed{0.53.}$

✓ Q4

5) compute the μ (mean) + SD(σ) of the value of the option

option value × probability

0(.47) + 1(.15) + 2(.13) + 3(.10) + 4(.10) + 5(0.5)

μ = 0 + .15 + .26 + .30 + .40 + .25

μ option value = 1.36

$\sigma = \sum (x - 1.36)^2 P$ = (0 − 1.36)²
= (−1.36)²
= 1.8496

1.8496 × 0.47 = 0.8693

0 × .47 = 0
1 × 0.15 = 0.15
2(0.13) = 0.26
3(0.10) = 0.30
4(0.10) = 0.40
5(0.05) = 0.25
+ 1.36